

# Haleigh E. Nyberg

NSF Graduate Research Fellow • Zonta Amelia Earhart Fellow

Ph.D. Candidate, Department of Earth, Atmospheric, and Planetary Sciences  
Purdue University

*Note: Publications prior to 2025 may appear under Haleigh E. Brown.*

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## EDUCATION

### Ph.D., Astrobiology

Aug 2023 – May 2028

Purdue University, Department of Earth, Atmospheric, and Planetary Sciences

Advisor: Dr. Stephanie Olson

Status: Ph.D. Candidate (ABD), advanced to candidacy April 2026

Dissertation Focus: Computational Modeling of Exoplanetary Habitability and the Origin of Life

Current GPA: 3.93/4.0

### B.A., Computational Physics (Summa Cum Laude)

Aug 2019 – May 2023

University of Montana Missoula, Department of Physics and Astronomy

Minors: Computer Science, Mathematics

Thesis: Advanced High-Altitude Balloon Trajectory Prediction

GPA: 3.91/4.0

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## ACADEMIC AWARDS AND FELLOWSHIPS

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|------------------------------------------------------------------------|-------------|
| ★ Zonta Amelia Earhart Fellowship                                      | 2026 – 2027 |
| ★ National Science Foundation Graduate Research Fellowship             | 2024 – 2029 |
| ★ Indiana NASA Space Grant Consortium Travel Grant (\$1,500)           | 2025        |
| ★ Ross Fellowship, Purdue Planetary Sciences Department                | 2023 – 2024 |
| ★ Dean's and President's List, University of Montana                   | 2019 – 2023 |
| ★ Montana University System Honors Scholarship (Full Tuition Coverage) | 2019 – 2023 |
| ★ Shallenberger Physics Scholarship, University of Montana             | 2019, 2022  |
| ★ John Pohl Musician's Scholarship, University of Montana              | 2019        |
| ★ Valedictorian, Sentinel High School                                  | 2019        |
| ★ Buddy DeFranco Improvisation Award                                   | 2018        |

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## RESEARCH EXPERIENCE

### Graduate Research Assistant

Aug 2023 – Present

Purdue University, Department of Earth, Atmospheric, and Planetary Sciences

Advisor: **Dr. Stephanie Olson**

- Run three-dimensional ExoPlaSim climate simulations across planetary obliquity, rotation rate, and stellar host to find which worlds sustain the wet-dry cycling that concentrates prebiotic chemistry toward an origin of life.
- Build pixel-wise machine learning models that predict ocean-current patterns from continental configuration, turning expensive circulation-model output into a fast surrogate for habitability screening.

### Lead Data Science Researcher

Jan 2025 – Dec 2025

Purdue University, The Data Mine

Corporate Partner: **Barrios Technology**

- Directed two 16-member, multi-university student teams as scrum master, building a dynamic Unreal Engine desktop and VR visualization tool for NASA's Lunar Gateway.

### Machine Learning Researcher

Aug 2024 – Dec 2024

Purdue University, The Data Mine

Corporate Partner: **John Deere**

- Built multivariate time-series forecasting models (ARIMA, LSTM) for John Deere demand planning, benchmarking methods to recommend a forecasting strategy they could put into production.

#### Undergraduate Research Intern and Peer Leader

May 2021 – Aug 2022

NASA Montana Space Grant Consortium, University of Montana

Advisor: **Dr. Jennifer Fowler**

- Cut balloon-launch training costs by \$283 per launch by adding dynamic drag-coefficient calculations to the trajectory-prediction software and rebuilding the campaign training procedure around it.

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#### PUBLICATIONS

- [2] Mingsuwan, C., Desano, L., Garcia, B., Agarwal, J., Ong, J., Sanchez, A., Xu, C., Cawthon, S., Lian, C., **Nyberg, H.E.** (2025). "Foundations of a Visualization Tool for NASA Gateway and Lunar Surface Operations." *The 6th Augmented, Virtual, and Mixed Realities (AR/VR) Workshop, IEEE Space Mission Challenges for Information Technology / Space Computing Conference (SMC-IT/SCC 2025)*. (Accepted).
- [1] **Brown, H.E.**, Capirala, A., Fifer, L. M., Laflèche, É. A., Nyberg, D., Payne, R., Sneed, E., Wong, M. L. (2025). "Strengthening Community Across Artists and Scientists in Astrobiology." *NASA DARES Request for Information*. (Accepted).

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#### ABSTRACTS AND CONFERENCE PRESENTATIONS

- [20] **Nyberg, H.E.**, Olson, S.L., Pearce, B.K.D., Bryant, R.N. (2026). Effects of Planetary Obliquity on Wet-Dry Cycles, Pond Organics, and Urability. *The Astrobiology Science Conference (AbSciCon) 2026*, Madison, WI, USA. 213-09. (**eLightning**)
- [19] **Nyberg, H.E.**, Capirala, A., Fifer, L. M., Laflèche, É. A., Nyberg, D., Payne, R., Sneed, E., Wong, M. L. (2026). PALLAS: Strengthening Community Across Artists and Scientists. *The Astrobiology Science Conference (AbSciCon) 2026*, Madison, WI, USA. 325-369. (**Poster Presentation**)
- [18] **Nyberg, H.E.**, Nyberg, D., Capirala, A., Fifer, L. M., Laflèche, É. A., Payne, R., Sneed, E., Wong, M. L. (2025). Planetary Obliquity and Origin of Life Potential in Warm Little Ponds: Earth and Beyond. *AGU Fall Meeting 2025*, New Orleans, LA, USA. (**Invited Speaker**)
- [17] **Nyberg, H.E.**, Nyberg, D., Capirala, A., Fifer, L. M., Laflèche, É. A., Payne, R., Sneed, E., Wong, M. L. (2025). PALLAS: Bridging the Art-Science Divide for Enhanced Scientific Communication. *AGU Fall Meeting 2025*, New Orleans, LA, USA. (**Poster Presentation**)
- [16] **Nyberg, H.E.**, Olson, S.L., Pearce, B.K.D., Bryant, R.N. (2025). Quantifying Planetary Origin of Life Potential: Connecting Global Climate to Local Wet Dry Cycling on Volcanic Islands. *Midwestern Geo-biology 2025 Conference*, West Lafayette, IN, USA. (**Oral Presentation**)
- [15] Capirala, A., Fifer, L.M., Laflèche, É.A., Nyberg, D., **Nyberg, H.E.**, Payne, R.C., Sneed, E.L., Wong, M.L. (2025). PALLAS: Building a Network of Artists & Scientists. *Goldschmidt 2025 Conference*, Prague, Czech Republic.
- [14] **Nyberg, H.E.**, Olson, S.L., Pearce, B.K.D., Bryant, R.N., Chavas, D.R. (2025). Planetary Obliquity Impacts Wet-Dry Cycling and the Potential for an Origin of Life in Warm Little Ponds. *Goldschmidt 2025 Conference*, Prague, Czech Republic. (**Oral Presentation**)
- [13] Batra, K., Hammond, T., **Nyberg, H.E.** (2025). Climatic Impacts of Orbital and Surface Processes on Exoplanet Habitability. *Goldschmidt 2025 Conference*, Prague, Czech Republic.
- [12] **Brown, H.E.**, Olson, S.L., Pearce, B.K.D., Chavas, D.R. (2024). Elucidating Planetary Scenarios That May Lead to an Origin of Life in Warm Little Ponds. *AGU Fall Meeting 2024*, Washington D.C., USA. P11D-3004. (**Poster Presentation**)
- [11] Calatayud, S., Capirala, A., **Brown, H.E.**, Jernigan, J., Olson, S.L. (2024). Exploring the Effects of Diverse Continental Configurations on Climate and Habitability. *AGU Fall Meeting 2024*, Washington D.C., USA. P11D-2999.
- [10] **Brown, H.E.**, Olson, S.L., Pearce, B.K.D., Bryant, R.N., Chavas, D.R. (2024). Planetary features of Earth-Like Exoplanets: Implications for an Origin of Life in Warm Little Ponds. *Exoplanets In*

*Our Backyard III*, New Haven, CT, USA. (**Poster Presentation**)

- [9] **Brown, H.E.**, Olson, S.L., Pearce, B.K.D., Bryant, R.N., Chavas, D.R. (2024). Planetary Features and a Wet-Dry Cycling Origin of Life. *Astrobiology Graduate Conference (AbGradCon) 2024*, Ithaca, NY, USA. (**Oral Presentation**)
- [8] Olson, S.L., Capirala, A., **Brown, H.E.** (2024). Continents Shape the Spatiotemporal Patterns of Marine Oxygen. *The Astrobiology Science Conference (AbSciCon) 2024*, Providence, RI, USA. 224-031.
- [7] **Brown, H.E.**, Chavas, D.R., Pearce, B.K.D., Olson, S.L. (2024). Impacts of Planetary Obliquity on Wet-Dry Cycling: Early Earth and Beyond. *The Astrobiology Science Conference (AbSciCon) 2024*, Providence, RI, USA. 325-035. (**Poster Presentation**)
- [6] Olson, S.L., Jernigan, J., Laflèche, É.A., **Brown, H.E.** (2024). Exploring Origin of Life Chemistry and Exoplanet Biosignatures with GCMs. *EGU General Assembly 2024*, Vienna, Austria. EGU24-14161.
- [5] **Brown, H.E.** (2023). Balloon Trajectory Prediction: Improving Calculation-based Techniques. *Senior Seminar (PHSX 499)*, University of Montana Department of Physics and Astronomy, Missoula, MT, USA. (**Oral Presentation**)
- [4] **Brown, H.E.**, Jetton, S., Hovenkotter, R., Rugg, H., Spangrude, C. (2022). Improving Calculation-based Balloon Trajectory Prediction. *Montana Space Grant Consortium Summer Research Symposium*, Missoula Public Library, Missoula, MT, USA. (**Seminar**)
- [3] Jetton, S., **Brown, H.E.**, Hovenkotter, R., Rugg, H., Spangrude, C. (2022). Balloon Trajectory Prediction: Improving Calculation-based Techniques. *AGU Fall Meeting 2022*, Chicago, IL, USA. P55D-1604. (**Poster Presentation**)
- [2] **Brown, H.E.**, Jetton, S., Hovenkotter, R., Rugg, H., Spangrude, C. (2021). Eclipse Campaign Simulation: Affordable Radiosonde Launches Using the Phantom 4. *Montana Space Grant Consortium Summer Research Symposium*, Missoula Public Library, Missoula, MT, USA. (**Seminar**)
- [1] **Brown, H.E.**, Reece, K., Spangrude, C., Covitt, B., Bernards, M., Fowler, J. (2021). Radiosonde Eclipse Campaign Education: Increasing Accessibility Through Adaptability and Affordability. *AGU Fall Meeting 2021*, New Orleans, LA, USA. ED15E-0558. (**Poster Presentation**)

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## TEACHING AND MENTORING EXPERIENCE

### Graduate Mentor, Amelia Earhart Leadership for Space Careers

Aug 2024 – Aug 2025

Purdue University, Honors College

- Facilitated discussions between senior space-industry professionals and undergraduate students on women’s leadership in space exploration and science, both remotely and in person through visits to NASA sites including Johnson Space Center.

### Fellowship Writing Coach

Aug 2024 – Present

Purdue University, Office of the Vice Provost for Graduate Students and Postdoctoral Scholars (OGSPS)

- Coach graduate and postdoctoral researchers across disciplines in securing fellowship funding (NSF GRFP, NDSEG, NIH F31/F32, Hertz, Ford Foundation), advising more than 40 applicants one-on-one on application strategy and grant writing.
- Advisees include a 2026 NSF Graduate Research Fellowship recipient, along with nominees for the Apple Scholars and Google PhD Fellowship programs.
- Trained and onboarded an incoming fellowship writing coach, shadowing her first review sessions to expand the program’s coaching capacity.

### Invited Funding Workshop Facilitator

Aug 2024 – Present

Purdue University, Office of the Vice Provost for Graduate Students and Postdoctoral Scholars (OGSPS)

- Designed more than a dozen distinct fellowship and funding workshops, including a three-part NSF GRFP series alongside dedicated NDSEG, NIH, SROP, Bridge, and computational-sciences funding sessions, and delivered them more than 80 times across six Purdue trainee and fellowship programs since 2024.

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| <b>Research Mentor for Silene Calatayud (Undergraduate)</b>                                                                              | Jan 2024 – May 2025 |
| Purdue University, Department of Earth, Atmospheric, and Planetary Sciences                                                              |                     |
| – Mentored in astrobiology research methodologies, co-authoring an AGU 2024 abstract with my mentee.                                     |                     |
| <b>Physics Lab Teacher (PHSX 205N, PHSX 206N)</b>                                                                                        | Jan 2022 – Jan 2023 |
| University of Montana, Department of Physics and Astronomy                                                                               |                     |
| – Independently instructed introductory physics lab sessions (College Physics I and II) for up to 24 undergraduate students per session. |                     |

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## LEADERSHIP AND SERVICE

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| <b>Session Co-convenor, Goldschmidt 2026 Conference</b>                                                                            | Jul 2026           |
| Session Title: <i>Chemistry and Climate of Exoplanets</i>                                                                          |                    |
| <b>Peer Reviewer, Planetary Science Journal (AAS)</b>                                                                              | 2026               |
| <b>Application Reviewer, Google PhD Fellowship Program</b>                                                                         | 2025, 2026         |
| Purdue nominee selection, reviewing alongside faculty panelists.                                                                   |                    |
| <b>Application Reviewer, Purdue Graduate Grant Competitions</b>                                                                    | 2026               |
| A.H. Ismail Interdisciplinary Graduate Grant and Frederick N. Andrews Environmental Travel Grant.                                  |                    |
| <b>Reviewer, Purdue Postdoctoral Travel Award</b>                                                                                  | 2025               |
| Scored applicant submissions across two funding cycles.                                                                            |                    |
| <b>Founding Treasurer, Purdue Golden Z Club</b>                                                                                    | Aug 2025 – Present |
| Co-founded Purdue’s chapter of Zonta International’s student service organization.                                                 |                    |
| Co-organized the Zonta Says NO! to Violence Against Women awareness walk (Nov 2025).                                               |                    |
| <b>Session Co-convenor, Goldschmidt 2025 Conference</b>                                                                            | Jul 2025           |
| Session Title: <i>Chemistry and Climate of Exoplanets</i>                                                                          |                    |
| <b>Executive Secretary, NASA Exoplanets Research Program (XRP) Review Panel</b>                                                    | 2026               |
| Coordinated panel logistics and review synthesis (Panel Week 3).                                                                   |                    |
| <b>Executive Secretary, NASA Habitable Worlds Program Review Panels</b>                                                            | 2024, 2025         |
| Recorded panel discussions and synthesized reviewer consensus for each proposal.                                                   |                    |
| <b>Co-Founder / Website Developer, PALLAS</b>                                                                                      | May 2024 – Present |
| PortAL Linking Artists and Scientists. pallas.gallery                                                                              |                    |
| – Designed and built the organization’s website and visual identity.                                                               |                    |
| – Helped secure an AAS Education and Professional Development mini-grant for a two-speaker pilot program (2025).                   |                    |
| – Co-organized “Envisioning Joint Futures in Art and Astronomy” (Apr 2026) and the AbSciCon 2026 Art and Science Mixer (May 2026). |                    |

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## PROFESSIONAL MEMBERSHIPS

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|---------------------------------------------------------------------------|------------------------|
| <b>Student Member, American Geophysical Union (AGU)</b>                   | 2021, 2022, 2024, 2025 |
| <b>Member, American Association for the Advancement of Science (AAAS)</b> | 2026 – 2027            |
| Student member; includes a subscription to <i>Science</i> .               |                        |

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## HIGHLIGHTED SKILLS

- ★ **Programming Languages:** Python (NumPy, SciPy, Pandas, Matplotlib, Seaborn, TensorFlow, PyTorch, Keras, scikit-learn, xarray), R, Java, MATLAB, Fortran
- ★ **Data Science and Machine Learning:** General Circulation Model Analysis, Convolutional Neural Networks, Bayesian Statistics, Multivariate Time Series Forecasting (ARIMA, LSTM), Principal Component Analysis, Logistic/Linear Regression, Random Forest, k-means Clustering, Gradient Descent, Algorithm Design, Data Structures, Data Mining, Data Visualization.
- ★ **Climate and Planetary Modeling:** ExoPlaSim, WRF, PlaSim; GCM data analysis (NetCDF) atmospheric dynamics, radiative transfer.
- ★ **Software and Technologies:** Git/GitHub, Linux/Unix, Jupyter Notebooks/Lab, Overleaf (LaTeX), SLURM/HPC, Unreal Engine, AutoCAD, Microsoft Office Suite, Google Workspace.

- ★ **Professional Skills:** Agile/Sprint Project Management, Scientific Writing and Presentation, Grant Proposal Writing, Interdisciplinary Collaboration, Peer Review, Public Speaking, Curriculum Development, Mentoring.
  - ★ **Certifications:** Responsible Conduct of Research (RCR), CITI Program, 2024; Research Security (NSPM-33), 2025; Data Science Sprint, The Data Mine, 2025; Data Handling and FERPA, 2024 to 2025.
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